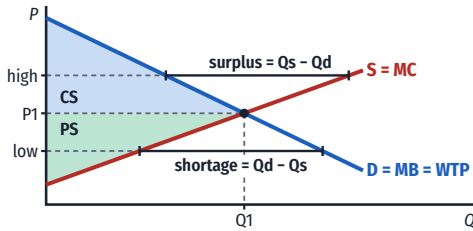


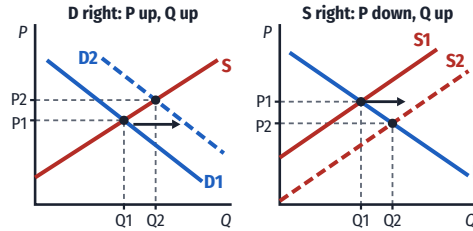
1 Market equilibrium and surplus



Label: $D = MB = WTP$; $S = MC$; P_1 , Q_1 ; CS; PS; surplus, shortage
 1. Draw D and S ; mark the crossing P_1 , Q_1 .
 2. Shade CS under D above P_1 , and PS above S below P_1 .

- $CS = \frac{1}{2} \times Q_1 \times (D \text{ intercept} - P_1)$. $PS = \frac{1}{2} \times Q_1 \times (P_1 - S \text{ intercept})$.
- Total surplus = $CS + PS$ is largest at Q_1 (no DWL).
- Above P_1 : surplus, P falls. Below P_1 : shortage, P rises.

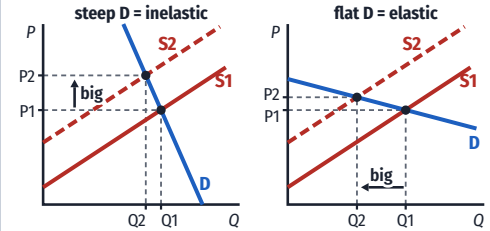
2 Shifting demand and supply



Label: D_1 to D_2 or S_1 to S_2 with an arrow; P_1 , Q_1 and P_2 , Q_2
 1. Buyer side (income, tastes) moves D ; cost side moves S .
 2. Shift it with an arrow; mark the new crossing P_2 , Q_2 .

- D right: P up, Q up. D left: P down, Q down.
- S right: P down, Q up. S left: P up, Q down.
- D right, S left: P up, Q unclear. Both right: Q up, P unclear.
- The good's own price moves along a curve, never shifts it.

3 Elasticity: steep versus flat demand

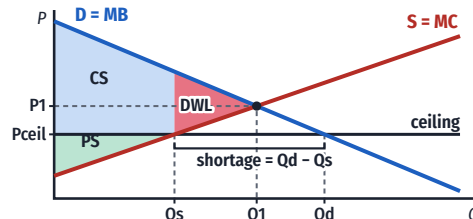


Label: steep = inelastic; flat = elastic; S_1 , S_2 ; P_1 , Q_1 , P_2 , Q_2

1. Draw the same S_1 and S_2 on both graphs.
2. Step D : P moves a lot, Q a little. Flat D : the reverse.

- $|\epsilon| = |\% \Delta Q \div \% \Delta P| = |1 / \text{slope}| \times (P / Q)$. $|\epsilon| > 1$ elastic.
- $|\epsilon| < 1$ inelastic. Vertical D : $\epsilon = 0$. Horizontal D : infinite.
- Spending $P \times Q$ rises with P if inelastic, falls if elastic.

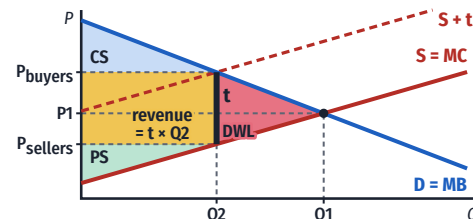
4 Price ceiling (and floor)



Label: $D = MB = WTP$, $S = MC$, P_1 , Q_1 , ceiling, Q_s , Q_d , shortage
 1. Draw D and S crossing at P_1 , Q_1 . Neither curve shifts.
 2. Flat ceiling below P_1 . Read Q_s on S and Q_d on D there.
 3. Shade DWL from Q_s to Q_1 . CS and PS stop at Q_s .

- Binds only below P_1 . Traded $Q = Q_s$. Shortage = $Q_d - Q_s$.
- $DWL = \frac{1}{2} \times (Q_1 - Q_s) \times (\text{height of } D \text{ at } Q_s - \text{ceiling})$.
- CS may rise (lower price) or fall (fewer units). PS falls.
- Floor binds only above P_1 : surplus = $Q_s - Q_d$, traded $Q = Q_d$.

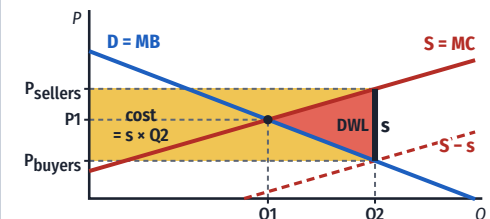
5 Per-unit tax



Label: $D = MB$, $S = MC$, $S + t$, P_1 , Q_1 , Q_2 , wedge t , P_{buyers} , P_{sellors}
 1. Draw D and S at P_1 , Q_1 , then $S + t$: S moved up by t .
 2. Q_2 is where $S + t$ meets D (wedge t between D and S).
 3. At Q_2 read P_{buyers} on D , then $P_{\text{sellors}} = P_{\text{buyers}} - t$ on S .
 4. Shade revenue box (0 to Q_2), DWL from Q_2 to Q_1 , CS, PS.

- Revenue = $t \times Q_2$. $DWL = \frac{1}{2} \times t \times (Q_1 - Q_2)$.
- Same result if buyers legally pay (draw $D - t$).
- Steeper side bears more: buyers $P_{\text{buyers}} - P_1$, sellers $P_1 - P_{\text{sellors}}$.

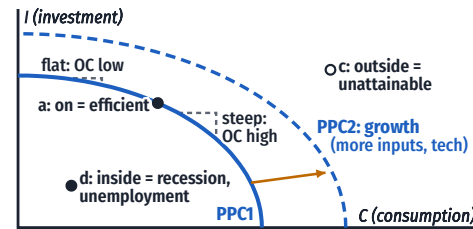
6 Subsidy



Label: $D = MB$, $S = MC$, $S - s$, P_1 , Q_1 , Q_2 , wedge s , P_{buyers} , P_{sellors}
 1. Draw D and S at P_1 , Q_1 , then $S - s$: S moved down by s .
 2. Q_2 is where $S - s$ meets D , so $Q_2 > Q_1$.

3. At Q_2 read P_{buyers} on D , then $P_{\text{sellors}} = P_{\text{buyers}} + s$ on S .
 4. Shade cost box (0 to Q_2) and DWL triangle from Q_1 to Q_2 .
- Cost = $s \times Q_2$. $DWL = \frac{1}{2} \times s \times (Q_2 - Q_1)$.
 - DWL because units from Q_1 to Q_2 have $MC > MB$.
 - CS gain + PS gain = cost - DWL.

7 PPC (production possibilities curve)

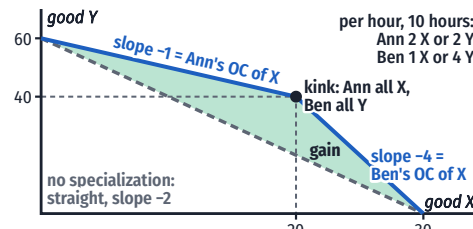


Label: axes C and I ; PPC1 bowed out; points a (on), d (inside), c (outside); PPC2 = growth

1. Bowed-out PPC1 between the two intercepts; mark a , d , c .
2. Growth (more labor, capital, technology): PPC2 outside.

- Slope = $-OC$ of the horizontal good. Bowed out = rising OC .
- Inside = recession, not an inward shift. More I today = bigger PPC later.

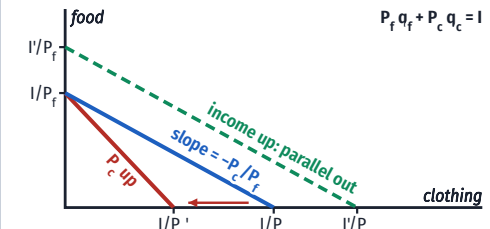
8 Comparative advantage: joint PPC



Label: intercepts; dashed line; kink; slopes = OCs; gain area
 1. Intercepts = sum of maxes (60 Y, 30 X); join with dashes.
 2. From 60 Y, slope -1 (Ann, lower OC of X) to the kink (20, 40), then slope -4 (Ben) to 30 X.

- OC of $X = Y$ per hour = X per hour (Ann 1, Ben 4). Ann, lower OC , makes X .
- Three or more people: add them in order of OC of X , lowest first; equal OC s share one segment.

9 Budget line and spending rule

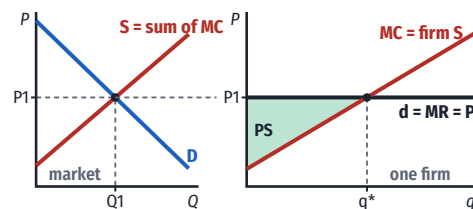


Label: food vertical, clothing horizontal; intercepts I/P_f , I/P_c ($I = \text{income}$); slope $-P_c/P_f$

1. Join $I \div P_f$ (food axis) and $I \div P_c$ (clothing axis).
2. Income up: parallel out. P_c up: pivot in on clothing.

- Rational Spending Rule: $MU_a \div P_a = MU_b \div P_b$.
- P_c up: $MU_c \div P_c$ falls below $MU_f \div P_f$, so buy less clothing, maybe more food, until equal.

10 Competitive firm and market

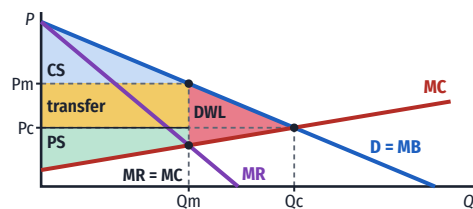


Label: market: D , $S = \text{sum of firm } MC$, P_1 , Q_1 ; firm: flat $d = MR = P$ at P_1 , $MC = \text{firm supply}$, q^* , PS

1. Market: D and S cross at P_1 . Carry P_1 across: flat $d = MR = P$.
2. Firm: rising MC ; q^* where MC crosses P ; shade PS.

- $MR = P$, so produce where $P = MC$. Firm supply = MC curve.
- Industry $S = \text{horizontal sum}$: add all firms' q at each P .
- Enter if PS area > fixed cost. Profit = $PS - \text{fixed cost}$.

11 Monopoly

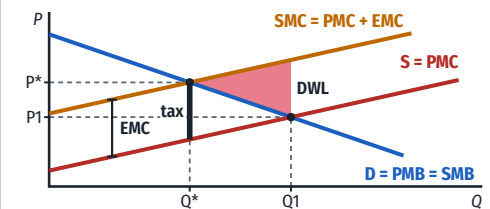


Label: $D = MB$; MR (same intercept, twice as steep, hits Q axis halfway); MC ; Q_m , P_m , Q_c , P_c ; CS, transfer, PS, DWL

1. Draw D , MR from the same intercept, rising MC .
2. Q_m where $MR = MC$; go straight up to D for P_m .
3. Q_c , P_c (competitive): D meets MC . DWL Q_m to Q_c .

- $P = a - bQ$ gives $MR = a - 2bQ$. No supply curve.
- Profit = transfer + PS = revenue - area under MC .
- Transfer: the CS rectangle between P_c and P_m becomes profit.
- Given Q_m and P_m : demand intercept $a = 2 \times P_m - MC$, slope $b = (P_m - MC) \div Q_m$.

12 Negative externality and Pigouvian tax



Label: $D = PMB = SMB$; $S = PMC$; $SMC = PMC + EMC$, EMC gap; Q_1 at $D = PMC$; Q^* at $D = SMC$; DWL; tax

1. D and $S = PMC$ cross at Q_1 . SMC sits EMC above S .
2. Q^* where SMC meets D . DWL from Q^* to Q_1 points at Q^* .

- Pollution above 0 is optimal: early units have $SMB > SMC$.
- $DWL = \frac{1}{2} \times EMC \times (Q_1 - Q^*)$ if EMC is constant.
- Pigouvian tax = EMC at Q^* moves the market to Q^* .
- Positive: $SMB = PMB + EMB$ above D ; too little made; subsidy = EMB at Q^* .

Col 1: grading words, OC, college, PPC, who specializes, shifters, P and Q after shifts, MU and spending rule ·
Col 2: elasticity, surplus, DWL, price controls, tax incidence · **Col 3:** firms, monopoly, pollution, number recipes

How to answer (his grading)

Inside the box, 2 to 6 sentences. "Explain" = give the reason.

- Use his words: MB, MC, MU, WTP, OC. $D = MB = WTP$. $S = MC$.
- Graphs: label both axes (P up, Q across) and every curve. Dotted lines to both axes, with values when known.
- Justify slopes: few substitutes means steep D; low, slowly rising MC means low, flat S.
- Unclear question: write how you read it, then answer that.

Question	Words his key requires
Cost of college	tuition plus foregone earnings
D slopes down	MB (MU, WTP) falls as Q rises
S slopes up	MC rises as Q rises
What PS is	P received minus each unit's MC
Who specializes	lower OC (comparative advantage)
Optimal pollution	SMC = SMB , usually Q above 0
Monopoly	MR = MC , then P read off D

Opportunity cost and the PPC

graph 7

OC = the value of the next best alternative you give up.

- Time: hours times your wage. Owned item: resale value now, not what you paid. Sunk costs: ignore.
- College = tuition, fees, books + **foregone earnings**. Rent and food cancel (paid either way).
- \$200 ticket plus 3 hours at a \$36 wage: OC = \$308.

PPC slope = minus the OC of the horizontal-axis good.

On the PPC	efficient
Inside	unused resources (recession); no shift
Outside	unattainable today
Bowed out	OC rises: lowest-OC producers switch first

Comparative advantage

graph 8

- OC of 1 X = (Y per hour) / (X per hour), for each person.

Lower OC = comparative advantage: specialize in it. Absolute advantage (more per hour) does not decide.

- Same % gain in one good for everyone, or in both goods for one person: OCs keep their order, so nothing changes.
- Equal OCs: no gains from specializing.
- Trade price lies between the two OCs, so both gain.
- "Always" or "solely" claims are usually false. Reasons not to specialize fully: national security, infant industries, different environmental or labor laws.

Supply and demand

graphs 1, 2

D right if	S right (down) if
income up (normal) or down (inferior)	input prices fall
a substitute's price rises	technology improves
a complement's price falls	more sellers
tastes move toward it	expect lower future P
buyers expect higher future P	tax or regulation cut
more buyers	

- The reverse of any row shifts that curve left.

Own price change = movement along the curve (Qd or Qs). Anything else shifts it.

- Buyer news: D. Seller news: S. Price control: neither.
- Both shift: D right + S left: P up, Q depends on the bigger shift. Both right: Q up, P depends.
- P and Q move together: D shifted. Opposite ways: S.
- One curve shifts: the steeper the other curve, the more P moves and the less Q moves. Other curve vertical (seats, planted crops): only P moves. Horizontal: only Q moves.
- Market D = horizontal sum of buyers' Qd at each P (it kinks where a buyer drops out). Labor market: firms demand, workers supply, wage = P.

Households

graph 9

- MU = change in total utility from one more unit. Total utility peaks where MU reaches 0.

Rational Spending Rule: $MU_a \div P_a = MU_b \div P_b$.

- MU falls with each extra unit, so demand slopes down.
- P_a rises: buy less a (MU_a rises), maybe more b (MU_b falls), until equal.

A change in MU at every Q (heatwave, taste) shifts demand. Own price change: move along it.

- Log utility: $MU = 1 \div c$. Two people's combined utility peaks when their MUs (so c) are equal: moving \$1 from richer to poorer raises it.

Elasticity

graph 3

$\epsilon = \% \Delta Q \div \% \Delta P$, with $\Delta = (\text{new} - \text{old}) \div \text{old}$.

- $\epsilon = (1 \div \text{slope}) \times (P \div Q)$, where slope = $\Delta P \div \Delta Q$.
- Demand ϵ is negative; judge size: -0.5 is inelastic (0.5 < 1).
- Spending: $\% \Delta (P \times Q) \approx \% \Delta P + \% \Delta Q$ for small changes; for big ones compute $P \times Q$ before and after.

ϵ	Curve	Spending if P up 1% (about)
-0.75	steep, inelastic	+0.25%
-1	unit elastic	0
-1.5	flat, elastic	-0.5%
-3	flatter, elastic	-2%

- Reverse use: $\% \Delta P = \% \Delta Q \div \epsilon$.
- Inelastic: no close substitute, a necessity, the short run. Elastic: close substitute, easy to do without, the long run, a narrow good (one brand).
- Supply: inelastic in the short run or with a fixed resource (oil wells, planted crops); elastic in the long run.
- A straight demand line is unit elastic at its midpoint: elastic above it, inelastic below.

Inelastic = steep, elastic = flat.

Surplus, efficiency, DWL

graph 1

- CS = area under D (willingness to pay) and above the price.

PS = area above S (MC) and below the price.

- Efficient = most total surplus: produce to $MB = MC$, highest-MB buyers buy, lowest-MC sellers sell.

DWL triangle points toward the efficient Q and grows out to the actual Q. Too little or too much Q both cause it.

- $DWL = \frac{1}{2} \times (Q \text{ gap}) \times (MB - MC \text{ gap at the actual } Q)$.
- First Welfare Theorem: competitive equilibrium with no market failure maximizes total surplus. It ignores equity: willingness to pay depends on income.

Price controls

graph 4

- A ceiling binds only below P_1 , a floor only above. No curve shifts.

Q traded = the smaller of Q_s and Q_d at the controlled P.

- One seller pricing below market with a purchase limit acts like a ceiling on its shelves: shortage, rationed by the limit, not by WTP.

Total surplus falls. CS can rise or fall: it gains a rectangle from PS , loses a triangle.

- Ceiling misallocation: units may not reach the highest-MB buyers, a loss on top of the DWL.
- Model misses fairness (gouging feels unfair) and market power.

Taxes and subsidies

graphs 5, 6

A tax t is a vertical wedge t between D and S at Q2: buyers pay P_{buyers} on D, sellers keep $P_{\text{buyers}} - t$ on S.

- Statutory incidence (who pays the government) does not matter: a tax on sellers shifts S up by t, on buyers D down by t. Economic incidence = who bears it.

The more inelastic (steeper) side bears more.

Flat S or vertical D	buyers pay all
Vertical S or flat D	sellers pay all

- Buyers' price rises by less than t unless S is flat or D vertical.
- DWL is bigger when D or S is elastic, zero if either is vertical.
- A tax in one small area: sellers can sell elsewhere, so supply is flat there and buyers bear it.
- No misallocation: price still rations.
- Real pass-through can be uneven (a Finnish sales tax cut was about half passed on, a rise nearly all). The model is a guide, not gospel.
- Price against tax across places: slope 1 = buyers bear all; slope 0 = sellers bear all; between = shared; above 1 = something else differs too (richer places have higher costs and taxes).

Competitive firms

graph 10

- Profit = revenue - total cost (counting the OC of every input).
- Price taker: too small to move P, so it faces flat demand and $MR = P$.

Produce where $MR = MC$, so $P = MC$. Firm supply = its MC curve.

- MC counts only costs that rise with one more unit (materials, hourly labor). Land, buildings, rent are fixed: they change profit, not MC or Q, and block entry in the short run, much less over years.
- **TRAP** Mistake: profit = PS. Correct: profit = PS - fixed cost. Enter only if PS (area between P and MC) > fixed cost.
- Input price rises: every MC up, S left (most for sellers who use the input most). Better technology: MC down, S right. Entry: S right.
- Free entry: profit draws entrants (S right, P falls) until profit = 0. Losses cause exit.

Monopoly

graph 11

- Market power: few close substitutes. Barriers to entry: high fixed costs, patents, licenses.
- $MR < P$: to sell one more unit it must cut P on every unit, losing revenue on units already sold.

1. Qm where $MR = MC$. 2. Go up from Qm to demand and read Pm.

- **TRAP** Mistake: P where MR meets MC. Correct: read P from demand, not from MC.
- Constant MC = c: $Q_m = (a - c) \div (2b)$, $P_m = (a + c) \div 2$, competitive $Q = 2 \times Q_m$.
- Patents: invention has a big fixed cost and low MC, so copies would push P to MC. A temporary monopoly pays for invention, with DWL until it expires.
- Policy: antitrust law (targets dominance and practices like paying to be the default, not only prices), price regulation, government ownership, patent limits.

Externalities

graph 12

- externality: a cost or benefit that falls on people outside the market and is left out of the price.

SMC = PMC + EMC, SMB = PMB + EMB. Market Q1: $D = S$. Efficient Q*: $SMB = SMC$.

	Negative	Positive
Market Q1	too much	too little
Fix	tax = EMC at Q^*	subsidy = EMB at Q^*

- External cost = area between SMC and PMC out to Q. Total surplus = CS + PS - external cost (+ tax revenue).
- Monopoly plus pollution: compare Q_m ($MR = PMC$) with Q^* ($D = SMC$). The monopoly cut can offset overproduction; landing exactly on Q^* is luck.
- **TRAP** Mistake: optimal pollution is zero. Correct: Q^* is where $SMC = SMB$; early units have $SMB > SMC$.
- Coase: clear property rights plus cheap bargaining let the parties fix it privately.
- Cap and trade: cap permits at Q^* , firms trade them (permit price = EMC at Q^*), so the cheapest cuts happen first.

Number problems with straight lines

Task	How (Q1 before, Q2 after)
Equilibrium	Demand P = supply P. Solve Q, then P.
CS	$\frac{1}{2} \times Q \times (\text{demand intercept} - P)$
PS	$\frac{1}{2} \times Q \times (P - \text{supply intercept})$
Tax t	Demand P = supply P + t. Revenue $t \times Q_2$. $DWL \frac{1}{2} \times t \times (Q_1 - Q_2)$.
Subsidy s	Demand P + s = supply P. Cost $s \times Q_2$. $DWL \frac{1}{2} \times s \times (Q_2 - Q_1)$.
Ceiling	Q traded = Q_s . Shortage = $Q_d - Q_s$. CS is a trapezoid: $\frac{1}{2} \times (\text{top} + \text{bottom height}) \times Q_s$.
Monopoly	$a - 2bQ = MC$ gives Q. P from demand.
Externality	Demand = SMC gives Q^* . $DWL \frac{1}{2} \times EMC \times (Q_1 - Q^*)$.

Check: $CS + PS + \text{tax revenue} + DWL = \text{original total surplus}$. Subsidy: $CS + PS - \text{cost} + DWL$.

- Example: $P = 10 - Q$ and $P = Q$ give $Q = P = 5$, $CS = PS = 12.5$. Tax 2: $Q = 4$, buyers pay 6, sellers keep 4, revenue 8, DWL 1.